

Marcelo D. Polêto

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Education

Ph.D. in Cell and Molecular Biology

UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL

Brazil

Mar. 2019

- **Dissertation Title:** Molecular flexibility of ligands as a design strategy of ligand-receptor interactions.

M.Sc. in Cell and Molecular Biology

UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL

Brazil

Jul. 2016

- **Thesis Title:** Parametrization strategies of aromatic rings commonly used in medicinal chemistry.

B.Sc. in Biochemistry

UNIVERSIDADE FEDERAL DE VIÇOSA

Brazil

Dez. 2014

- **Thesis Title:** Structural analysis of *Porcine circovirus 2* capsid protein: genetic diversity role on cellular adsorption and viral stability.
- Science Without Border Fellowship - Uppsala Universitet (2013-2014)

Employment and Experience

Research Associate - USP

SUPERVISOR: FERNANDO SEGATO

Lorena - SP, Brazil

Sept. 2024 - Present

Research Associate - Virginia Tech

SUPERVISOR: JUSTIN A. LEMKUL

Blacksburg - VA, USA

Nov. 2023 - Aug. 2024

Postdoctoral Associate - Virginia Tech

SUPERVISOR: JUSTIN A. LEMKUL

Blacksburg - VA, USA

Dez. 2020 - Nov. 2023

Postdoctoral Associate - Universidade Federal de Viçosa

SUPERVISOR: JULIANA L.R. FIETTO

Viçosa - MG, Brazil

Apr. 2019 - Dez. 2020

Skills

Programming Python, LaTeX, CHARMM, Matplotlib

Techniques Molecular Dynamics, Enhanced Sampling, Force field development, Quantum Chemistry

Softwares CHARMM, OpenMM, GROMACS, Plumed, Gaussian09, Psi4, PyMol, VMD

Peer-Reviewed Publications

1. Michel, H.M.; **Polêto, M.D.**; Lemkul, J.A. *Running Gaussian-accelerated Molecular Dynamics Simulations in NAMD [Article v0.1]*. Living Journal of Computational Molecular Science, 6(1), 3815, 2025.
2. **Polêto, M.D.**; Lemkul, J.A. *Structural and electronic properties of polyethylene terephthalate (PET) from polarizable molecular dynamics simulations*. Macromolecules, 58, 1, 403–414, 2025.
3. **Polêto, M.D.**; Allen, K.D.; Lemkul, J.A. *Structural dynamics of the methyl-coenzyme M reductase active site are influenced by coenzyme F₄₃₀ modifications*. Biochemistry, 63 (14), 1783–1794, 2024.
4. **Polêto, M.D.**; Lemkul, J.A. *Differences in Conformational Sampling and Intrinsic Electric Fields Drive Ion Binding in Telomeric and TERRA G-Quadruplexes*. Journal of Chemical Information and Modeling, 63 (21), 6851-6862, 2023.
5. Giacon, N., Lo Cascio, E., Davidson, D.S., **Polêto, M.D.**, Lemkul, J.A., Pennacchietti, V., Pagano, L., Zamparelli, C., Toto, A., Arcovito, A. *Structural and functional insights on the interaction between SARS-CoV-2 E protein and ZOI-*

6. **Polêto, M.D.**; Lemkul, J.A. *TUPÃ: Electric field analyses for molecular simulations*. Journal of Computational Chemistry, 43(16), 2022.
7. **Polêto, M.D.**; Lemkul, J.A. *Integration of experimental data and use of automated fitting methods in developing protein force fields*. Communications Chemistry, 5(1), 2022.
8. Arantes, P.R.; **Polêto, M.D.**; Pedebos, C.; Ligabue-Braun, R. *Making it rain: cloud-based molecular simulations for everyone*. Journal of Chemical Information and Modeling, 61(10), 2021.
9. De Oliveira, T.V.; **Polêto, M.D.**; Barbosa, S.V.; Coimbra, J.S.R.; De Oliveira, E.B. *Impacts of Ca²⁺ cation and temperature on bovine α -lactalbumin secondary structures and foamability – Insights from computational molecular dynamics*. Food Chemistry, 367(15), 2021.
10. De Oliveira, T.V.; Guimarães, A.P.; Bressan, G.C.; Maia, E.R.; Coimbra, J.S.R.; **Polêto, M.D.**; De Oliveira, E.B. *Structural and molecular bases of angiotensin-converting enzyme inhibition by bovine casein-derived peptides: an in silico molecular dynamics approach*. Journal of Biomolecular Structure and Dynamics, 39(4), p. 1386-1403, 2020.
11. Rusu, V.H.; Santos, D.E.S.; **Polêto, M.D.**; Galheigo, M.M.; Gomes, A.T.A.; Verli, H.; Soares, T.A.; Lins, R.D. *Rotational Profiler: A Fast, Automated, and Interactive Server to Derive Torsional Dihedral Potentials for Classical Molecular Simulations*. Journal of Chemical Information and Modeling, 60(12), p. 5923-5927, 2020
12. **Polêto, M.D.**; Grisci, B.I.; Dorn, M.; Verli, H. *ConfiD: an analytical method for conformational characterization of small molecules using molecular dynamics trajectories*. Bioinformatics, 36(11), p. 3576–3577, 2020.
13. Arantes, P.; Pedebos, C.; **Polêto, M.D.**; Pol-Fachin, L.; Verli, H. *The Lazy Life of Lipid-Linked Oligosaccharides in All Life Domains*. Journal of Chemical Information and Modeling, 2019 (online).
14. De Oliveira, A.S.; Gazolla, P.A.R.; Oliveira, A.F.C.S.; Pereira, W.L.; Viol, L.C.; Maia, A.F.S.; Santos, E.G.; Da Silva, I.E.P. ; Mendes, T.A.O.; Da Silva, A.M.; Dias, R.S.; Da Silva, C.C. ; **Polêto, M.D.**; Teixeira, R.R.; De Paula, S.O. *Discovery Of Novel West Nile Virus Protease Inhibitor Based On Isobenzonafuranone And Triazolic Derivatives Of Eugenol And Indan-1,3-Dione Scaffolds*. Plos One, v. 14, p. e0223017, 2019
15. **Polêto, M.D.**; Alves, M.P.; Ligabue-Braun, R.; Eller, M.R.; De Carvalho, A.F. *Role Of Structural Ions on the Dynamics of the Pseudomonas Fluorescens 07a Metalloprotease*. Food Chemistry, 286(15), p. 309-315, 2019.
16. Arantes, P.; **Polêto, M.D.**; John, E.B.; Pedebos, C.; Grisci, B.I.; Dorn, M.; Verli, H. *Development of Gromos-Compatible Parameter Set for Simulations of Chalcones and Flavonoids*. Journal Of Physical-Chemistry B, 123(5), p. 994-1008, 2019.
17. Tesch, R.; Becker, C.; Müller, M.P.; Beck, M.E.; Quambusch, L.; Getlik, M.; Lategahn, J.; Uhlenbrock, N.; Costa, F.N.; **Polêto, M.D.**; De Sena, P.M.; Rodrigues, D.A.; Sant'Anna, C.M.; Ferreira, F.F.; Verli, H.; Fraga, C.A.M.; Rauh, D. *An Unusual Intramolecular Halogen Bond Guides Conformational Selection*. Angewandte Chemie-Internacional Edition, v. 57(31), p. 9970-9975, 2018.
18. **Polêto, M.D.**; Rusu, V.H. ; Grisci, B.I. ; Dorn, M.; Lins, R.D.; Verli, H. *Aromatic Rings Commonly Used In Medicinal Chemistry: Force Fields Comparison And Interactions With Water Toward The Design Of New Chemical Entities*. Frontiers In Pharmacology , v. 9, p. 395, 2018.
19. Fischer, N.M.; **Polêto, M.D.**; Steuer, J.; Van Der Spoel, D. *Influence of Na⁺ And Mg²⁺ Ions on RNA Structures Studied With Molecular Dynamics Simulations*. Nucleic Acids Research (Online) , v. 46(10), p. 4872-4882, 2018
20. Lima, R.N.; Faheem, M.; Barbosa, J.A. R. G.; **Polêto, M.D.**; Verli, H.; Melo, F.L.; Resende, R.O. *Homology Modeling And Molecular Dynamics Provide Structural Insights Into Tospovirus Nucleoprotein*. BMC Bioinformatics , v. 17, p. 11-17, 2016.
21. Guimarães-Peixoto, R.P. M.; Pinto, P.S. A.; Santos, M. R.; **Polêto, Ma.D.**; Silva, L.F.; Silva-Júnior, A. *Evaluation Of a Synthetic Peptide From The Taenia Saginata 18kDA Surface/Secreted Oncospheral Adhesion Protein For Serological Diagnosis Of Bovine Cysticercosis*. Acta Tropica , v. 164, p. 463-468, 2016.
22. Figueira, F.; Farinha, A.S. F. ; Muteto, P.V. ; **Polêto, M.D.** ; Verli, H.; Gomes, M.T.S. R. ; Tomé, A.C. ; Cavaleiro, J.A. S. ; Tomé, J.P.C. *[28]Hexaphyrin Derivatives For Anion Recognition In Organic And Aqueous Media*. Chemical Communications (London. 1996. Print), v. 52, p. 2181-2184, 2016.
23. Ramos, C. ; Figueira, F.; **Polêto, M.D.** ; Amado, F.M.L ; Verli, H; Tomé, J.P.C ; Neves, M.G.P.M.S . *Esi-*Ms/Ms* Of Expanded Porphyrins: a Look Into Their Structure And Aromaticity*. Journal Of Mass Spectrometry (Print), v. 51,

p. 342-349, 2016.

24. Siqueira, R.P.; Barbosa, E.A.A.; **Polêto, M.D.**; Righetto, G.L.; Seraphim, T.V.; Salgado, R.L.; Ferreira, J.G.; Barros, M.V.A.; De Oliveira, L.L.; Laranjeira, A.B.A.; Almeida, M.R.; Júnior, A.S.; Fietto, J.L.R.; Kobarg, J.; De Oliveira, E.B.; Teixeira, R.R.; Borges, J.C.; Yunes, J.A.; Bressan, G.C. *Potential Antileukemia Effect And Structural Analyses Of SRPK Inhibition By N-(2-(Piperidin-1-Yl)-5-(Trifluoromethyl)Phenyl)Isonicotinamide (SRPIN340)*. Plos One, v. 10, p. e0134882, 2015.
25. Salgado, R.L.; Vidigal, P.M.P.; Gonzaga, N.F.; De Souza, L.F.L.; **Polêto, M.D.** ; Onofre, T.S.; Eller, M.R. ; Pereira, C.E.R.; Fietto, J.L.R.; Bressan, G.C. ; Guedes, R.M.C.; Almeida, M.R.; Silva Júnior, A. *A Porcine Circovirus-2 Mutant Isolated In Brazil Contains Low-Frequency Substitutions In Regions Of Immunoprotective Epitopes In The Capsid Protein*. Archives Of Virology, v. 160 (11), p. 2741-2748, 2015.

Journal Reviewer

- Journal of Chemical Information and Modeling
- Nature Communications Chemistry
- Journal of Chemical Theory and Computation
- Journal of Biomolecular Structure and Dynamics
- The Journal of Physical Chemistry

Fellowships

Research Support Foundation of Minas Gerais State (FAPEMIG)

UNDERGRAD RESEARCHER - 2010-2011

Viçosa - MG, Brazil

2010-2011

Research Support Foundation of Minas Gerais State (FAPEMIG)

UNDERGRAD RESEARCHER - 2011-2012

Viçosa - MG, Brazil

2011-2012

Research Support Foundation of Minas Gerais State (FAPEMIG)

UNDERGRAD RESEARCHER - 2012-2013

Viçosa - MG, Brazil

2012-2013

Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES)

SCIENCE WITHOUT BORDERS - 2013-2014

Uppsala, Sweden

2013-2014

Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES)

POSTDOCTORAL RESEARCHER - 2019-2020

Viçosa - MG, Brazil

2019-2020

Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP)

POSTDOCTORAL RESEARCHER - 2024-PRESENT

Lorena - SP, Brazil

2024-Present

Invited Seminars and Presentations

2024 CHARMM Developers Meeting

PRESENTER OF **OPTIMIZING PET BIORECYCLING SOLUTIONS THROUGH ELECTRONICALLY POLARIZABLE SIMULATIONS,**

Baltimore, USA

Jul 2024

52nd Annual Meeting of the Brazilian Society of Biochemistry and Molecular Biology (SBBq)

PRESENTER OF **ACTIVE-SITE ELECTRONIC PROPERTIES AND CONFORMATIONAL DYNAMICS OF PET-DEGRADING ENZYMES**

Aguas de Lindoia, BRA

Jun 2023

American Chemical Society Meeting

PRESENTER OF **MODULATION OF POLYETHYLENE TEREPHTHALATE (PET) ELECTRONIC PROPERTIES BY PET-DEGRADING ENZYMES**

Indianapolis, USA

March 2023

Biophysical Society 67th Annual Meeting

PRESENTER OF **ACTIVE-SITE ELECTRONIC PROPERTIES AND CONFORMATIONAL DYNAMICS OF PET-DEGRADING ENZYMES.**

San Diego, USA

February 2023

Brazilian Congress of Industrial Biochemistry

PRESENTER OF **COMPUTATIONAL BIOPHYSICS: PAST, PRESENT AND FUTURE**

Online, Brazil

Jan 29 2021

Proteomics and Protein Biochemistry Symposium

PRESENTER OF **COMPUTATIONAL BIOPHYSICS: PAST, PRESENT AND FUTURE**

Viçosa - MG, Brazil

Ago 14 2020

Nacional Pharmaceutical Sciences Congress

PRESENTER OF **DRUG DESIGN: WHERE WE ARE AND WHERE WE GO FROM HERE**

Online, Brazil

May 08 2020

Courses Taught

Biophysical characterization of protein-ligand complexes

CREDITS: 1

Viçosa - MG, Brazil

Spring 2019

BIO791 - Methods in Structural Bioinformatics

CREDITS: 3

Viçosa - MG, Brazil

Spring 2019

Theory and Practice of Molecular Dynamics Simulations

CREDITS: 1

Viçosa - MG, Brazil

Spring 2017 and Fall 2018

Bioinformatics for Biomedicine scholars

CREDITS (ONLY PRACTICES): 2

Porto Alegre - RS, Brazil

Spring 2017

Basic Bioinformatics

CREDITS (ONLY PRACTICES): 1

Porto Alegre - RS, Brazil

Spring 2017

Mentorship

Matheus Aragão

MENTOR

PhD in Industrial Biotechnology

2024 - Present

Haley M. Michel

MENTOR

PhD in Biochemistry

2020 - Present

Laura I. Gil-Pineda

MENTOR

PhD in Biochemistry

2020 - Present

Thomás V. de Oliveira

CO-ADVISOR

PhD in Food Technology

2017 - 2020

Marcel A. Diogo

CO-ADVISOR

M.Sc. Student in Applied Biochemistry

2019 - 2021

João Victor B. de Moraes

CO-ADVISOR

M.Sc. Student in Applied Biochemistry

2021 - Present

Júlia M. Ferreira

CO-ADVISOR

B.Sc. Student in Biochemistry

2019 - 2023

Isabela S. Gomes

CO-ADVISOR

B.Sc. in Chemical Engineering

2019

Gabriel Alves

CO-ADVISOR

M.Sc. in Plant Pathology

2023

Other Professional Services

“Recent Advances in Molecular Force Fields” symposium

PRESIDER

ACS National Meeting

Spring 2021

Softwares

TUPÃ - Electric field analyses

WEBSITE - [HTTPS://GITHUB.COM/MDPOLETO/TUPA/](https://github.com/mdpoletto/tupa/)

2022

ConfID - Conformational Identifier

WEBSITE - [HTTPS://GITHUB.COM/SBCBLAB/CONFID](https://github.com/SBCBLAB/CONFID)

2019

Technical Production

Making it Rain: cloud-based molecular simulations for everyone

MOLECULAR DYNAMICS TUTORIALS USING GOOGLE COLAB

2021